

RWANDA TVET BOARD
COMPETENCY-BASED ASSESSMENT
[FORMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 5 | **Total Marks:** 40

Module: Machine Learning Application | **Date:** 2026-05-25 | **Time:** 1 Hour 30 Minutes

Name: _____ Index No: _____ Class: _____

SECTION A: TRUE / FALSE

Answer True (T) or False (F).

- 1 Reinforcement Learning is irrelevant to developing Machine Learning Model. (1 marks)

Answer: **True** **False**

- 2 Input feature is irrelevant to developing Machine Learning Model. (1 marks)

Answer: **True** **False**

- 3 Support Vector Machine has no role when developing Machine Learning Model. (1 marks)

Answer: **True** **False**

- 4 K-Means clustering is a key concept when developing Machine Learning Model. (1 marks)

Answer: **True** **False**

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

- 5 What is the role of Supervised Learning when developing Machine Learning Model? (1 marks)

- A. Feature engineering
B. K-Means Clustering
C. Supervised Learning
D. Reinforcement Learning (RL)

Answer: _____

- 6 How is Unsupervised Learning used when developing Machine Learning Model? (1 marks)

- A. Data preparation
B. Unsupervised Learning
C. Computation power
D. Anomaly Detection

Answer: _____

- 7 How is Training dataset used when developing Machine Learning Model? (1 marks)

- A. Model training
B. Machine learning
C. ML Algorithm
D. Training dataset

Answer: _____

- 8 Which of the following correctly defines Cross-validation? (1 marks)

- A. Gathering a machine learning dataset

- B. Cross-validation
 - C. Flask API
 - D. Activation function
- Answer: _____

9 How is Linear Regression used when developing Machine Learning Model? (1 marks)

- A. Logistic Regression
 - B. Model Accuracy
 - C. Linear Regression
 - D. ML Algorithm
- Answer: _____

1 What is the role of Random Forest when developing Machine Learning Model? (1 marks)

- A. Python
 - B. Random Forest
 - C. Learning rate
 - D. Data preparation
- Answer: _____

1 What is the significance of Neural Network when developing Machine Learning Model? (1 marks)

- A. Python
 - B. Neural Network
 - C. Cross-validation
 - D. Each operation
- Answer: _____

SECTION C: MATCHING

Match each term in Column A with Column B.

1 Match each term in Column A with its correct description in Column B. (2 marks)

Column A (Terms)	Column B (Descriptions)
1. Supervised Learning	A. is the simplest machine learning model to understand in which
2. Linear regression	B. is the simplest machine learning model in which we try to predict one
3. The representation of linear regression	C. is a linear equation, which combines a set of
4. Logistic Regression	D. is a statistical model used to predict a binary outcome (i.e., 0 or 1,

Answers: 1.____ 2.____ 3.____ 4.____

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Answers: 1.____ 2.____ 3.____ 4.____

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

- 1 Analyze the importance of Machine Learning in developing Machine Learning Model. (1 marks)
4 Identify its advantages and limitations.

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- 1 Demonstrate how to use Logistic Regression when developing Machine Learning Model. (4 marks)
5 Provide a step-by-step explanation.

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- 1 Explain the role of K-Nearest Neighbors when developing Machine Learning Model. (4 marks)
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- 1 Evaluate the effectiveness of Deep Learning when developing Machine Learning Model. (4 marks)
7 Justify your answer with examples.

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- 1 Define scikit-learn library and explain its importance in developing Machine Learning (4 marks)
8 Model.

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- 1 Demonstrate how to use NumPy array when developing Machine Learning Model. (4 marks)
9 Provide a step-by-step explanation.

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- 2 Evaluate the effectiveness of Data preprocessing when developing Machine Learning (4 marks)
0 Model. Justify your answer with examples.

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